ZENFLOW: Investigating MR Transitions for Enhancing Sleep and Relaxation

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Figure 1: An overview of three relaxation methods. (left) ZENFLOW without space transformation (ZF): user sees virtual elements overlay on their physical environment, (middle) Pranayama, a traditional breath-control technique, and (right) Zenflow with space transformation (ZF_{ST}): user sees their physical space is transformed (ceiling opens to sky) in addition to overlay of virtual elements.

Abstract

Stress and poor sleep remain pervasive challenges in modern life, yet traditional relaxation practices such as pranayama (breathing exercises) require guidance, discipline, and environments that are often difficult to sustain. VR–based relaxation tools have emerged as alternatives, but their abrupt immersion into fully virtual environments can feel disruptive and misaligned with the gradual nature of meditative practices. To address this gap, we collaborated with pranayama practitioners in a co-design process to develop ZENFLOW, an MR system that blends subtle visuals and breathing cues to gradually transform the user's surroundings into a restorative virtual space. We evaluated the system in a 3 week within-subjects study (N=12), comparing traditional Pranayama with two variations of ZENFLOW. Results show that ZENFLOW transition design significantly improved self-reported sleep quality and objective measures of stress and sleep. Our work contributes design insights and evidence that gradual environmental transition can improve MR systems for stress management.

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CCS Concepts

• **Human-centered computing** → **User studies**; Visualization design and evaluation methods; **Mixed / augmented reality**.

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1 Introduction

Stress is one of the most pressing health challenges in contemporary life, linked to impaired cognition, weakened immunity, cardiovascular risk, and disrupted sleep [64]. While traditional approaches such as meditation, breathwork, and cognitive-behavioral therapy are effective [19, 27, 50], they often require sustained practice, dedicated environments, and access to trained professionals [15].

Meditation practices calm the body's stress system by lowering cortisol and reducing over-activation of the HPA (Hypothalamic Pituitary Adrenal) axis. Slow, deep breathing activates the parasympathetic nervous system, helping to slow heart rate, reduce blood pressure, and steady attention [52, 73]. However, traditional methods of practice have their limitations. Meditation, for example, can feel repetitive, with limited opportunity for novelty, making it difficult to sustain engagement and often leading to dropout outside structured settings [37, 40].

Also, deep breathing exercises typically require quiet, dedicated spaces that are difficult to find in busy or stressful environments. Standard practices such as Cognitive Behavioral Therapy, while effective, often show dropout rates over 25 percent due to time constraints, emotional fatigue, and limited access to trained professionals [14, 42]. For individuals already coping with fatigue, anxiety, or chronic health conditions, sustaining these practices becomes even harder. These barriers highlight the gap between proven clinical efficacy and real-world accessibility.

Technology-mediated relaxation techniques have emerged as a promising alternative to make evidence-based techniques more accessible [34, 46, 65]. Mobile apps make guided mindfulness and breathing exercises available anytime, reducing reliance on trained professionals [36]. Building on this accessibility, immersive systems such as the Meta Quest extend these benefits by offering embodied environments that significantly lower heart rate and perceived stress in short-term trials[32, 55], by creating focused, distraction-free experiences [12, 39, 49]. Studies have demonstrated that embodied VR interactions, such as breathing-controlled visuals or meditative multisensory spaces [56], deepen presence and support emotional regulation beyond traditional approaches [20, 69, 71]. These platforms provide opportunities to deliver powerful traditional methods in novel, embodied ways. However, most immersive relaxation applications are designed by technologists alone. Co-design with experts in traditional relaxation practices (e.g., clinical psychologists, meditation teachers) remains limited, despite growing recognition that participatory design processes are essential for developing effective therapeutic XR interventions [45, 53].

In this work, we address this gap by engaging mental health professionals who are practitioners of pranayama (traditional breathing exercise) [29, 62] in the co-design of an immersive relaxation system [59]. From this process, we derived five key design insights including the importance of *environmental transition*. Whereas abrupt shifts into fully virtual environments can disrupt presence and add cognitive load [54, 72], gradual transformations could foster a more natural, embodied relaxation process.

Building on these insights, we developed ZENFLOW, an MR system that progressively blends the user's physical surroundings into a virtual night-sky environment. This transition is paired with subtle breathing guidance and a minimalist, non-anthropomorphic agent. Unlike prior systems that focus on content, ZENFLOW emphasizes seamless perceptual continuity as a core design principle.

We evaluated two variations of ZENFLOW (with and without environmental transition) in a three-week within-subjects study (N=12), comparing both against the traditional Pranayama method [29, 62] (Fig. 1). Both self-reported measures (stress and sleep questionnaires) and wearable data (Garmin-derived stress and sleep scores) revealed that ZENFLOW's environmental transition led to significantly higher relaxation and better sleep outcomes. In summary, this paper makes three contributions:

- We contribute design insights from a co-design process with pranayama practitioners to inform immersive relaxation design.
- We present ZENFLOW, a novel MR-based technique that progressively transforms physical space into a calming virtual environment.
- We provide empirical evidence that environmental transition methods are critical determinants of relaxation efficacy in immersive systems.

2 Related Work

The pursuit of effective relaxation techniques has led to a diverse landscape of solutions, ranging from traditional practices to immersive technologies. Traditional methods such as meditation and breathwork emphasize self-regulation, while audio and mobile applications broaden accessibility through guided content. Immersive approaches, from flotation tanks to VR, offer novel ways of reducing sensory load or transporting users into calming environments, yet often struggle with accessibility, integration into daily life, or over-occlusion of the physical world. Recent work in wearable XR and biofeedback points to new opportunities for personalization and embodiment, but the design of transitions between realities—and the therapeutic value of involving clinical experts in shaping them—remains underexplored. In the following sections, we review these four strands of prior work and position our contribution within this evolving landscape.

2.1 Mind–Body Practices

Traditional methods such as meditation, breathwork, and Pranayama breathing exercises have long been used to regulate stress and support sleep. These practices effectively modulate autonomic responses [63], reduce perceived stress [15], and improve cardiovascular outcomes. However, they often require disciplined self-practice and environments free of distraction, which limits their accessibility and consistency in everyday life.

Practices like Pranayama, though highly effective in regulating the autonomic nervous system and promoting relaxation [63], require sustained commitment and dedicated practice to achieve deeper states of calm. This motivates exploration of more accessible and technology-supported approaches to relaxation.

2.2 Digital and Mobile Relaxation Tools

Building on traditional practices, digital solutions such as guided meditations, binaural beats, and ambient soundscapes have become widely adopted. Their portability and low barrier to entry make them attractive, and studies have shown efficacy in reducing anxiety [8]. Mobile applications (e.g., Headspace[1], Calm[2]) have further democratized access by offering structured programs, progress tracking, and guided exercises. Meta-analytic findings show moderate improvements in stress, anxiety, and depression [75], while randomized controlled trials report significant stress reduction among novice meditators using app-based interventions [74].

While convenient, these methods remain limited by screen-based distractions and a lack of deep immersion, pointing to opportunities for more embodied and immersive technologies.

2.3 Immersive Environments for Relaxation

Beyond conventional methods, dedicated immersive environments have been developed to facilitate deep relaxation. Flotation therapy (or sensory deprivation tanks) offers profound relaxation by minimizing external sensory input, leading to altered states of consciousness and significant stress reduction [44]. While highly effective, these solutions are typically non-portable, expensive, and require specialized facilities, limiting their widespread accessibility.

As VR technology has become more affordable and widespread, it has enabled entirely new approaches to relaxation [13, 43]. VR applications transport users to serene environments (e.g., tranquil beaches, calming forests), providing a strong sense of presence that distracts from real-world stressors and induces relaxation [25, 26]. VR's strength lies in its ability to deliver fully controlled and customizable experiences [10], but its complete occlusion of the physical world can lead to disorientation and feelings of detachment, limiting suitability for prolonged use.

Recent CHI work on blended realities highlights the importance of aligning physical and virtual spaces [2] to maintain continuity and presence [21, 22]. These findings suggest that not only the content, but also the method of transitioning into immersive experiences, is a critical determinant of their effectiveness.

2.4 Wearable XR for Relaxation and Well-being

Wearable headsets and physiological sensors are increasingly being used in XR-based stress management. Studies have investigated the use of VR head-mounted displays for pain management [20], anxiety reduction in clinical settings [6, 67, 70], and mindfulness training [1, 48, 51]. Similarly, early work in AR has explored overlaying calming visual or auditory elements onto the real world to enhance mindfulness during daily activities [30, 66, 68]. Some research has also looked into integrating physiological sensing (e.g., heart rate, skin conductance) [58] from wearables with immersive experiences to provide biofeedback-driven relaxation [4, 23, 57]. One pilot study found that AR-guided meditations significantly reduced heart rate and skin conductance levels during and after sessions, compared to baseline measurements [30]. Another systematic review of AR interventions confirmed their efficacy in reducing anxiety, particularly in phobia exposure and general well-being, although sample sizes were often small [3]. These approaches offer increased accessibility and the advantage of practicing relaxation exercises without detaching from one's current environment. While these studies demonstrate the potential of wearable immersive technologies, many existing AR/VR relaxation applications tend to either fully replace the real environment or simply overlay digital content without a sophisticated integration strategy that considers the user's perception of transition between realities [18].

[1] https://www.headspace.com/

[2] https://www.calm.com/

2.5 Participatory and Therapeutic Design Approaches

A recurring critique in XR for well-being is the lack of clinician involvement. Fewer than 12% of therapeutic VR/AR/MR systems report integrating mental health experts early in development [45]. Yet participatory design methods are essential for ensuring that interventions align with therapeutic needs and avoid unintended harm [53]. While some systems have involved users in prototyping, few explicitly co-design immersive relaxation experiences with clinical experts, leaving a gap in both content appropriateness and transition design choices. Beyond therapeutic contexts, prior work on co-design in immersive systems has highlighted how process awareness and appropriate representational tools can empower participants to actively shape XR experiences [11, 24]. Such insights underscore the importance of carefully structuring co-design sessions with clinical experts to ensure both participation and therapeutic appropriateness.

2.6 Summary and Research Gap

Building upon these foundations, our system introduces a novel approach to immersive relaxation by focusing on the progressive and seamless transition from the user's real environment into an augmented virtual space and back. Unlike traditional VR, which often relies on abrupt environmental shifts, or conventional AR, which primarily overlays content, our MR experience is designed to gradually transform the user's surroundings. This unique methodology addresses a critical gap in current immersive relaxation research: the impact of the method of environmental transition on user experience and relaxation efficacy. By carefully orchestrating this merger of virtual elements in reality space, our system aims to minimize cognitive load, enhance presence, and foster a more natural and profound state of relaxation. Our evaluation, correlating self-reported relaxation with objective physiological data from a Garmin Venu 3s, provides empirical evidence that this progressive transition is not only preferred by participants but also leads to measurably improved sleep scores and reduced stress levels. This highlights the importance of design considerations beyond just content, emphasizing the how of immersion as a key factor in developing highly effective and user-centric relaxation applications for wearable headsets.

3 Co-Design with Practitioners

3.1 Participants

We conducted three 90-minute co-design workshops with five professionals who practice pranayama, including two clinical psychologists, one neuropsychologist, and two counseling psychologists. Their average clinical experience was 8.4 years, and they specialized in areas such as stress, trauma, and sleep disorders. Details of the co-design participants - their country, professional experience, and clinical expertise in stress, sleep, and breathwork - are summarized in Table 1.

3.2 Procedure

The sessions followed a semi-structured format that introduced different immersive technologies (VR, AR, MR), shared early concept sketches, played sample animations, and invited open discussions on visuals, sound, interaction methods, and comfort. Health experts shared their input through verbal feedback, sketches, and by using a shared Miro board. Details of the workshop structure are summarized in Table 2.

Environment: The first session focused on exploring an appropriate environment for the experience. Experts discussed various nature-based options like forests and sunrises. However, during the discussion, they raised the following concerns: forest scenes could feel overwhelming or trigger anxiety in trauma patients (*"The enclosed canopy could activate claustrophobic responses in trauma patients"* - E1), and sunrise imagery might remind users of work or daily responsibilities, causing stress *"Patients report sunrise imagery reminds them of impending work demands"* - E3). After considering these factors, they agreed that a starry night sky would be the most calming and suitable environment.

Virtual Guide: The second session centered on the design of the virtual guide. Experts felt that using a human-like avatar might feel too personal or distracting, especially for users dealing with social anxiety. Instead, they suggested a more abstract, neutral presence-like a soft glowing orb with a gentle, non-commanding voice (*"Human-like faces might trigger social evaluation anxiety in individuals experiencing depression"* - E2).

Breath–Visual Synchrony: In the final session, we refined the timing between breath and visual feedback. To illustrate the concept, we developed a basic MR prototype that visualized breathing in real time. Experts observed that when the visuals did not closely align with the user's breathing, the experience could feel distracting rather than calming. They recommended keeping the delay under 200 milliseconds to maintain a smooth and intuitive experience. During this phase, experts made some interesting remarks. For example, E2 suggested that *"Patients often feel more relaxed at night because that's when cortisol levels naturally drop. Moonlight helps create a calming atmosphere that makes it easier for them to let go of overthinking and mental noise"*. Similarly, E5 emphasized that *"The virtual guide should function like a peripheral nervous system extension - present but never commanding"*. In addition, E1 highlighted that *"Breathing visuals must mirror diaphragmatic rhythm to enhance interoceptive awareness without cognitive effort""*.

3.3 Design Insights

Here we summarize the design features that emerged from our co-design process.

3.4 D1: Relaxation Pathways for MR Design

We identified two core relaxation pathways that serve as the foundation for MR intervention design:

- Body awareness pathways (activated through breath focus) that help users tune into physical sensations.
- Distraction pathways (engaged via immersive scenes) that redirect attention away from stressors

These pathways formed the conceptual framework for our modalities:

- Breath-Sync MR builds body awareness for better stress recognition
- Celestial MR shifts focus away from discomfort through cosmic visuals
- Music MR uses rhythm to help release physical tension

These distinct mechanisms highlight the importance of purposefully selecting therapeutic modalities based on targeted neurophysiological outcomes (See Table 3).

3.5 D2: Ambient Environment Design

Through the co-design process, experts emphasized that not all natural scenes are equally calming for patients. In the first workshop, forest environments were considered but quickly dismissed as potentially overwhelming or even anxiety-inducing for individuals with trauma histories (*"The enclosed canopy could activate claustrophobic responses in trauma patients"* - E1). Similarly, sunrise imagery was identified as problematic, as it often reminded patients of daily responsibilities and work pressures rather than encouraging relaxation (*"Patients report sunrise imagery reminds them of impending work demands"* - E3).

The discussion led the group to select the night sky as the environment most likely to promote relaxation. The experts noted that starry skies tend to feel gentle on the eyes, culturally comforting, and universally associated with calm and wonder. One expert highlighted that patients "often feel more relaxed at night because cortisol levels naturally drop," and the presence of moonlight helps them let go of mental noise (E2). Other experts added that starry skies tend to feel gentle on the eyes and their cultural associations with calm and wonder made them a universally safe and soothing choice. Importantly, this decision emerged directly from the practitioner's clinical experience during co-design, and it is further supported by prior research showing that celestial environments evoke feelings of awe and reduce stress levels [7]. Figure 2 illustrates the designed MR environment, showing the celestial night-sky backdrop and the minimalist guide character developed during the co-design process.

Table 1: Details of the Pranayama Practitioners part of the Co-Design Stage

ID	Country	Experience	Clinical Background and Expertise
E1	United Kingdom	6 years	Clinical psychology researcher and practicing consultant; pranayama practitioner. Focus on anxiety disorders and integration of breathwork in therapy.
E2	United Arab Emirates	5 years	Clinical psychologist and pranayama practitioner. Specializes in sleep disorders and mindfulness-based interventions for stress relief.
E3	Canada	6 years	Counselling psychologist and pranayama practitioner. Expertise in workplace stress, sleep, and overall well-being programs.
E4	India	7 years	Neuropsychologist and pranayama practitioner. Works on stress- and anxiety-related neurocognitive issues and regulation strategies.
E5	India	5 years	Practicing psychologist and pranayama practitioner. Focus on stress management, sleep hygiene, and general psychological well-being.

Table 2: Co-Design Workshop Structure

Phase	Activity	Focus	Output
Session 1: Concept Exploration	MR demo + symptom vignettes	Identify stress triggers and coping mechanisms	Refined design goals, user personas, contextual use cases
Session 2: Sensory Prototyping	Miro board environment sketching	*"Forests may trigger agoraphobia" (E1)*	Night sky selected over sunrise/forest
Session 3: Agent Interaction	Figma walkthrough of virtual guide	*"Avoid anthropomorphism to prevent uncanny valley" (E3)*	Luminous orb with gender-neutral voice

Table 3: Design Foundations for MR Modalities

Modality	Core Approach	Key Benefit	Designer Insight
Celestial MR	Attention shift	Reduces tension	*"Starry scenes calm threat responses" (E2)*
Breath-Sync MR	Body awareness	Improves stress recognition	*"Breath-visual sync boosts awareness" (E4)*
Music MR	Rhythm matching	Breaks *stress cycles*	*"Theta rhythms help release tension" (E1)*

Note. Practitioners' inputs: E1, E2, E4 refer to collaborators. Quotes represent verbatim insights from co-design sessions.

3.6 D3: Agent Design

We avoid human-like guides to prevent participants from feeling unsettling or culturally biased. Instead, the virtual helper appears as a simple glowing orb that pulses gently in sync with the user's breathing. It only whispers briefly in a neutral voice at lower volume. This simple design follows the expert's advice that virtual guides should function as "autonomic nervous system extensions", quietly supporting the body's natural calming processes - rather than giving commands. This makes users feel less judged while keeping the support helpful.

3.7 D4: Breathing Guidance

During the early design phase, we explored different ways to visually guide the user's breathing. One of them involved showing a mouth-like animation on the virtual guide to mimic breathing. However, the team decided not to use this approach, as it could distract users, feel too human-like, and make the experience feel less calming.

Instead, a simple and clean design was chosen: a soft blue line that slowly expands and contracts, matching a calm breathing rhythm. This acts as a gentle guide, helping users follow along with their breathing at their own pace without needing instructions. The lines maintain their movement regardless of the user's actual breathing

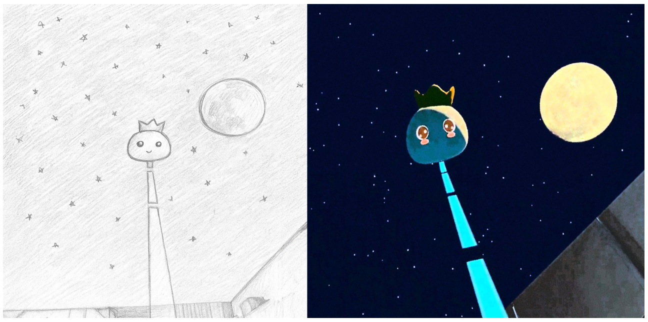


Figure 2: Pencil sketch (left) and digital sketch (right) illustrating the MR environment and agent design, including the celestial backdrop, moon, and simplified guide character.

pattern, offering consistent guidance without drawing attention to mismatches. The key technical and clinical design parameters for this implementation are summarized in Table 4.

The visuals were optimized to respond quickly and smoothly, ensuring the animation feels natural and uninterrupted. By keeping the guidance subtle and avoiding any human-like figures, the system supports relaxation and self-awareness without making users feel like they have to perform or follow strict rules.

3.8 D5: Sound Design

During design sessions, experts advised against using common nature sounds, such as rain, birds, or forests, as they might remind some users of past experiences or trigger unwanted emotions, especially in those with trauma backgrounds. *"Avoid recognizable sounds that might trigger memories. Neutral tones help users stay present in their body."* (E3)

Instead, a soundscore was chosen that uses gentle, neutral tones without strong cultural or emotional associations. These included soft, crystal-like sounds within a calming frequency range, selected to help users relax without evoking personal memories. Additionally, melodies or strong beats were avoided so that the sound remains in the background and doesn't distract users, allowing them to focus on their breath and the present moment.

3.9 D6: Safety

Through co-design, we established sensory boundaries to prevent discomfort. For example, sound changes capped at 30dB, brightness is kept at a minimum (10% of original brightness of pass-through for the most part) to avoid triggering anxiety. Also it was suggested to use clinically meaningful metaphors that make abstract concepts tangible, like moonlight symbolizing 'switching off' worried thoughts.

4 ZENFLOW System Design

Our system integrates an immersive MR application with physiological data collection to evaluate relaxation outcomes. Figure 3 shows the overall architecture, including the MR rendering pipeline, and interaction logic.

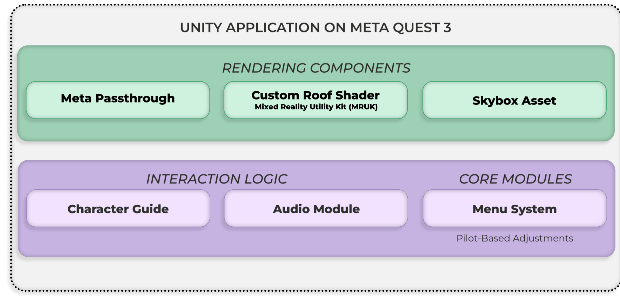


Figure 3: System Architecture that illustrates the rendering components of the MR (ZF_{ST}) application and the interaction logic that was designed during the codesign phase.

4.1 Space Transformation

To enable seamless environmental transitions, ZENFLOW utilizes Meta's Mixed Reality Utility Kit (MRUK) to scan and interpret the user's physical space. During initialization, the system attempts to identify key architectural features—particularly the ceiling. If the ceiling is not detected, the application prompts a room scan to ensure accurate spatial mapping.

Once the ceiling dimensions are obtained, a custom shader—adapted from MRUK's base shader—is applied to the roof component. This shader gradually reveals a virtual skybox, starting from the center and expanding outward in a smooth, time-bound animation. The transition is triggered by a specific sequence within the breathing guidance module, ensuring synchronization between visual transformation and relaxation cues.

To maintain immersion and realism, the virtual guide adheres to the Scene API, allowing for dynamic occlusion handling. This ensures that if any physical object obstructs the line of sight between the user and the guide, the occlusion is rendered appropriately, preserving spatial coherence and enhancing perceptual continuity.

4.2 Application Development

The application was developed in Unity 2022.3 LTS and deployed on the Meta Quest 3 headset, chosen for its robust passthrough capabilities and accessibility, which enable seamless blending of real and virtual environments. To support MR experiences, we utilized Meta's Passthrough API[3]. During setup, users completed a room calibration process, allowing the system to capture spatial details of their physical environment. This step was essential for accurately anchoring virtual elements and ensuring realistic environmental transitions. We include a minimal virtual guide, following prior research that emphasizes simplicity to avoid distraction. Its design was intentionally kept abstract and unobtrusive to maintain focus on relaxation. For MR transitions, we implemented a custom shader that progressively replaced the real ceiling with a dynamic starry skybox. This gradual reveal was central to achieving a smooth, non-disruptive transition into the virtual environment. Figure 4 illustrates this workflow, showing the guide's appearance in the

[3]https://developers.meta.com/horizon/documentation/unity/unity-passthrough/

Table 4: Breathing Guidance Design

Design Feature	User Benefit	Implementation
Simple Visuals	Less distraction, calming	Soft blue lines expand/contract slowly
Always Visible	No frustration	Lines soften but stay visible if out of sync
Fast Response	Feels natural	Updates in < 150ms (Unity Timeline)
No Instructions	Self-guided breathing	No text or voice prompts
Non-Human Form	Comfortable experience	No face/mouth animations

physical space and the smooth transition of the ceiling into a starry night-sky environment.

The user interface was designed for simplicity, featuring a menu button for session control. Each relaxation session lasted approximately eight minutes, a duration informed by prior work on optimal meditation timing. Audio cues were sourced from a validated relaxation library to ensure quality and consistency. Additional features included timed triggers for the virtual guide, gradual room darkening, and an optional breathing guide that was dynamically disabled during certain phases based on pilot feedback to support natural, self-paced breathing.

5 User Study

5.1 Participants and Recruitment

Participants were recruited from a pool of office workers within the university community. This demographic was specifically targeted due to the inherent stress associated with their work environment, which aligns with the study's focus on relaxation. Inclusion criteria included specific age ranges and types of work to ensure a relatively homogenous sample relevant to the study's objectives. A total of 12 office workers (N=12, 7 male, 5 female; mean age = 30.7 ± 2.8 years) successfully completed all phases of the study. Participants reported diverse professional backgrounds within the university environment, consistent with our recruitment strategy targeting individuals experiencing work-related stress. All participants provided informed consent prior to their involvement.

5.2 Procedure

The study employed a within-subjects design with counterbalanced conditions to mitigate order effects. Each participant experienced all three relaxation modalities: traditional Pranayama and two versions of our ZENFLOW application (with and without space transformation). The entire study spanned three consecutive weeks for each participant, with a structured schedule:

5.2.1 Familiarization Session. Upon recruitment, each participant underwent a comprehensive familiarization session. This session was crucial for ensuring comfort and proficiency with the technology and study procedures. Key aspects covered included:

- **Headset Basics:** Participants were introduced to the fundamental operation of the Meta Quest 3 headset.
- **Room Environment Setup:** Detailed instructions and hands-on practice were provided for performing the initial room setup, including drawing physical boundaries within their personal space. This ensured the application could accurately map their environment for the ZF and ZF_{ST} experiences.
- **Application Navigation:** Participants were guided through the process of launching the study application and navigating its internal menus and features.
- **Wearable Device Setup:** The Garmin Venu 3s smartwatch was set up for each participant, and they were instructed on how to wear it correctly for continuous data collection over the subsequent three weeks.

5.2.2 Study Schedule and Condition Design: The intervention followed a structured three-week schedule. Each week was dedicated to a single relaxation condition, performed across five working days (Monday to Friday), with weekends serving as off-days to allow for washout and natural recovery. The three conditions Pranayama (PR), ZENFLOW without space transformation (ZF), and ZENFLOW with space transformation (ZF_{ST}) were randomized and counterbalanced across participants. For example, one participant might engage in PR during Week 1, ZF in Week 2, and ZF_{ST} in Week 3, while another followed a different order. Within each assigned week, participants were instructed to complete one session per day on the five working days, performed within 30 minutes before their intended sleep time. The immersive ZENFLOW sessions (ZF and ZF_{ST}) were approximately eight minutes in duration, consistent with the application design.

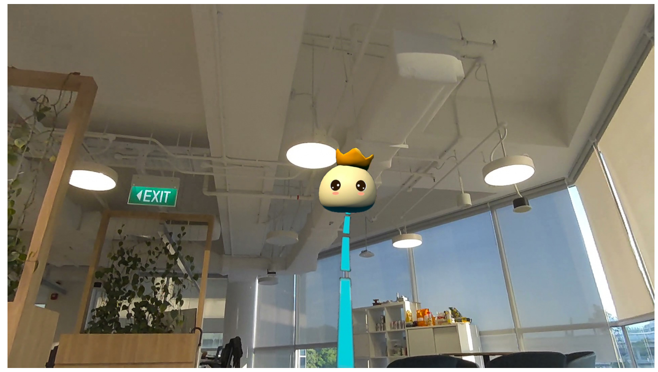
5.3 Data Collection and Analysis

We collected self-reported questionnaires (subjective data) as well as physiological data from a smartwatch (objective data) to study the impact of the different relaxation methods on stress and sleep. This dual approach helps holistic understanding of perceived effects as well as physiological changes in stress and sleep. We provide details below. Figure 5 provides an overview of the study design and data collection schedule, showing the three counterbalanced conditions across weeks and the daily flow of questionnaires and physiological recordings.

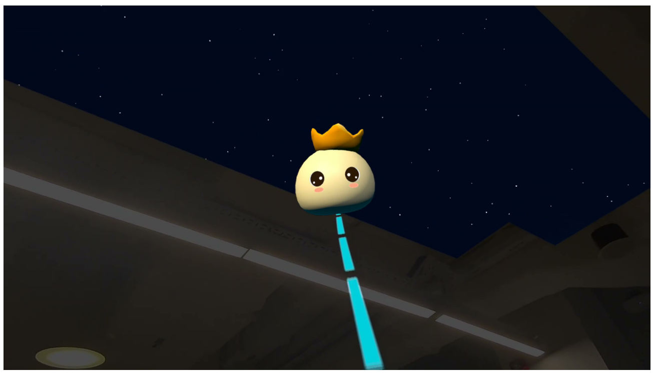
Self-Reported Questionnaire: We recorded two questionnaires. (1) To assess subjective stress, participants fill out the **Perceived Stress Scale (PSS-10)** [4] questionnaire [9], consisting of 10 items on a 5-point Likert scale, all of which are added together to obtain an overall stress score between 0 – 40 (higher score indicates higher perceived stress). Participants answered this questionnaire twice – before the relaxation experience, and after the experience (in the morning following sleep). (2) To assess subjective sleep, participants completed the **Leeds Sleep Evaluation Questionnaire (LSEQ)** [47] in the morning. LSEQ consists of four subscales – *GTS* (ease of Getting To Sleep), *QOS* (Quality Of Sleep), *AFS* (Awakening From

[4] We adapted the wording of each item from "in the last month" to "in the last day" to align with the daily study design.

(a) Guide appears; ambient lighting preserved.



(b) User engages with the guide -without spatial transition (ZF)



(c) (ZF_{ST}) Sky reveal begins; room darkens.

(d) ZENFLOW with transition of ceiling to night sky (ZF_{ST})

Figure 4: Screenshots of the ZENFLOW application: (a) and (b) show the experience without spatial transitions (ZF); (c) and (d) illustrate the spatial effects (ZF_{ST}), including ambient lighting changes and sky reveal.

Sleep), and *BFW* (Behavior Following Wakefulness), each rated on a 100-point scale (higher value indicates favourable outcomes) which we assessed individually in Section 5.5.

Physiological Data: We collected physiological data from Garmin Venu 3s smartwatch synced via the FitRockr API [16]. For objective stress, we recorded **Garmin-reported stress scores** during the 30 minutes prior to sleep (based on the recorded sleep onset time). Garmin quantifies stress on a 0–100 scale (higher values = higher stress), based on heart rate and heart rate variability [17]. To track objective sleep, we recorded **inter-beat interval (IBI)** readings, i.e., the time between successive heart rates, during the first 30 minutes of sleep. In general, IBI is an indicator of parasympathetic activity, where a higher value implies reduced physiological stress. One participant did not synchronize the device reliably and lacked complete watch recordings across intervention conditions; consequently, physiological analyses were restricted to participants with complete device data.

Data Analysis: We analyzed both self-reported questionnaire scores and physiological measures using linear mixed-effects models. Each model included *Intervention* (PR, ZF, and ZF_{ST}) as a fixed effect and a participant-specific random intercept to account for repeated measures. Models were fit by restricted maximum likelihood (REML) via `lme4` and accessed from Python using `pymer4` [5, 31].

For fixed effects, we obtained F-tests based on Type III sums of squares with Satterthwaite-approximated degrees of freedom as implemented in `lmerTest` [33, 60]. We computed omnibus tests and estimated marginal means (EMMs; a.k.a., least-squares means) with `emmeans`, using `joint_tests()` for ANOVA-like omnibus tests [35, 61]. Pairwise comparisons among Intervention levels were performed on the EMMs with single-step multiplicity control based on the multivariate-t distribution (`adjust="mvt"`), following the simultaneous-inference framework underlying `multcomp` [28, 35].

We report EMMs with 95% CIs and multiplicity-adjusted p-values. For visualization, we show boxplots of within-participant mean differences for the three comparisons (ZF − PR, ZF_{ST} − PR, and ZF_{ST} − ZF), making the repeated-measures structure explicit.

5.4 Debriefing and Reimbursement

Following the completion of all three weekly sessions, participants attended a final debriefing session. During this session, the study's full objectives were revealed, any remaining questions were addressed, and all study devices (Meta Quest 3 and Garmin Venu 3s) were collected. Participants were reimbursed for their time and effort upon successful completion of all study requirements.

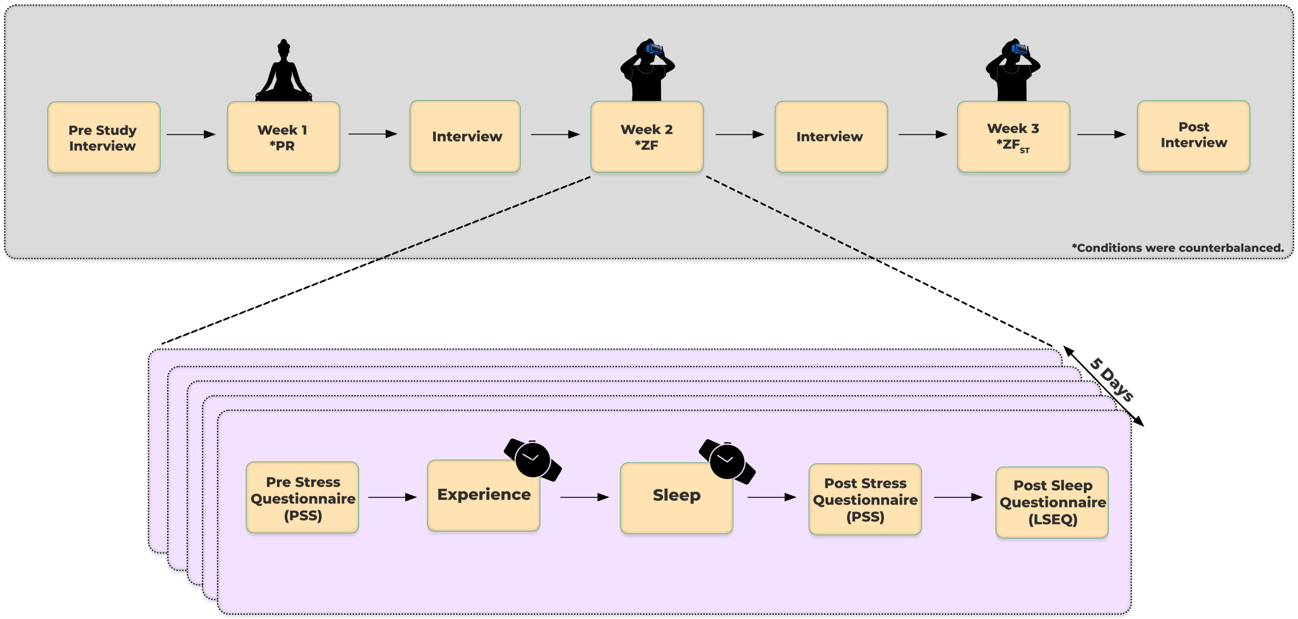


Figure 5: Overview of the three-week study protocol with randomized relaxation conditions (PR, ZF, ZF_{ST}), data collection, and analysis pipeline.

5.5 Results

5.5.1 Subjective Measures: Questionnaire Data. Self-reported data from the Perceived Stress Scale (PSS-10) and the Leeds Sleep Evaluation Questionnaire (LSEQ) provided crucial insights into participants' subjective experiences.

Subjective Sleep: We analyzed the four LSEQ subscales –*GTS*, *QOS*, *AFS*, and *BFW* – using linear mixed-effects models with *Intervention* (PR, ZF, ZF_{ST}) as a fixed effect and a random intercept for participant.

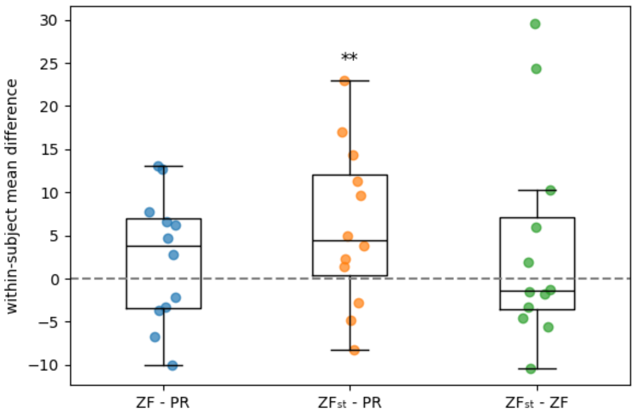
Significant main effects of *Intervention* were observed for all subscales ($F = 3.26$–8.80, $p < .05$). EMMs and pairwise contrasts indicated that the ZF_{ST} condition consistently outperformed PR across all subscales. For example, for *GTS*, ZF_{ST} improved scores relative to PR by $\Delta = +7.35$ (SE = 2.17, 95% CI [2.22, 12.48], $p = .0025$). Similarly, *QOS* showed a difference of $\Delta = +8.36$ (SE = 2.01, 95% CI [3.61, 13.11], $p < .001$). ZF also improved *QOS* and *BFW* compared to PR ($p < .02$), but differences between ZF_{ST} and ZF were smaller and not statistically significant after adjustment. Figure 6 illustrates within-subject differences. This indicates that ZENFLOW without spatial transition (ZF) alone improved some subscales compared to baseline. However, adding spatial transition ZF_{ST} significantly outperformed baseline (PR) across all subscales.

Perceived Stress: Analysis of morning PSS scores did not reveal statistically significant differences between conditions (all p > .40). Estimated mean differences were small (≤ 1 point on the PSS) and confidence intervals intersected with zero, suggesting comparable stress levels across the conditions. Similarly, changes in PSS across the day (evening–morning) were not significantly different between conditions (all p > .55). Nonetheless, a consistent but non-significant trend indicated that participants in the ZF_{ST} condition reported slightly lower stress than those in the ZF condition. While these effects were small in magnitude and not statistically significant, they suggest that ZF_{ST} may hold potential for reducing perceived stress relative to ZF, warranting further investigation with larger samples.

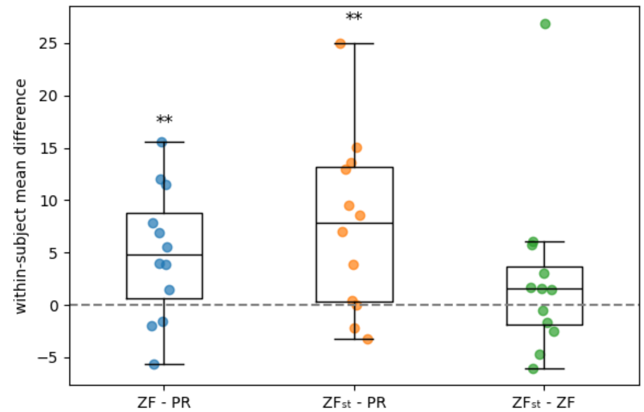
5.5.2 Objective Measures: Physiological Data. To complement the subjective measures, we analyzed physiological data from the Garmin smartwatch, focusing on stress and inter-beat interval (IBI) during the pre-sleep period and early sleep, to assess how each condition influenced autonomic recovery.

Objective Stress Scores. We analyzed the average Garmin stress scores during the 30 minutes preceding sleep onset across the three conditions. Results revealed a significant difference between PR and ZF_{ST}, with stress scores being lower in the ZF_{ST} condition ($\Delta = -6.62$, $SE = 2.60$, 95% CI $[-12.78, -0.45]$, $p = .032$). Comparisons between PR and ZF, as well as ZF_{ST} and ZF, did not reach statistical significance ($p = .85$ and $p = .10$, respectively), although the estimates suggest that ZF_{ST} tended to reduce pre-sleep stress relative to both alternatives. These findings indicate that ZF_{ST} may be more effective in lowering stress levels prior to sleep, complementing the subjective reports of relaxation.

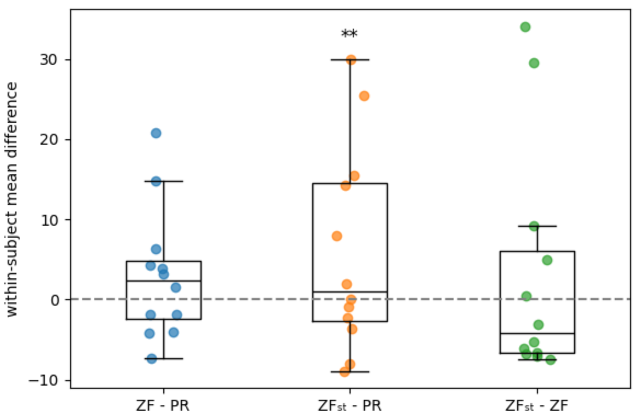
Objective Sleep Metric from IBI. We further examined IBI during the first 30 minutes of sleep as an indicator of parasympathetic activity. Raw IBI time series were preprocessed to correct artifacts using the Kubios automatic artifact-correction algorithm[38, 41], and the mean IBI was calculated. The analysis revealed significantly



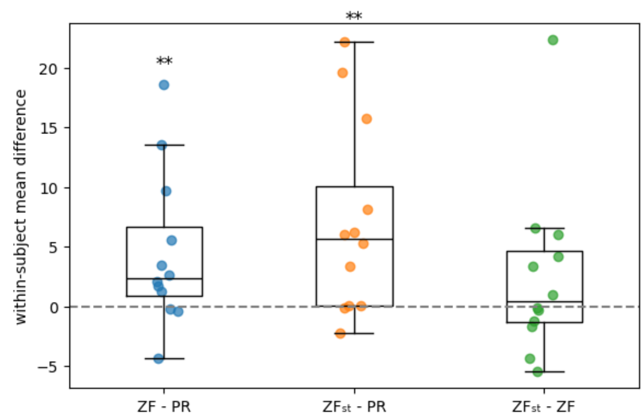
(a) GTS (ease of getting to sleep)



(b) QOS (quality of sleep)



(c) AFS (awakening from sleep)



(d) BFW (behavior following wakefulness)

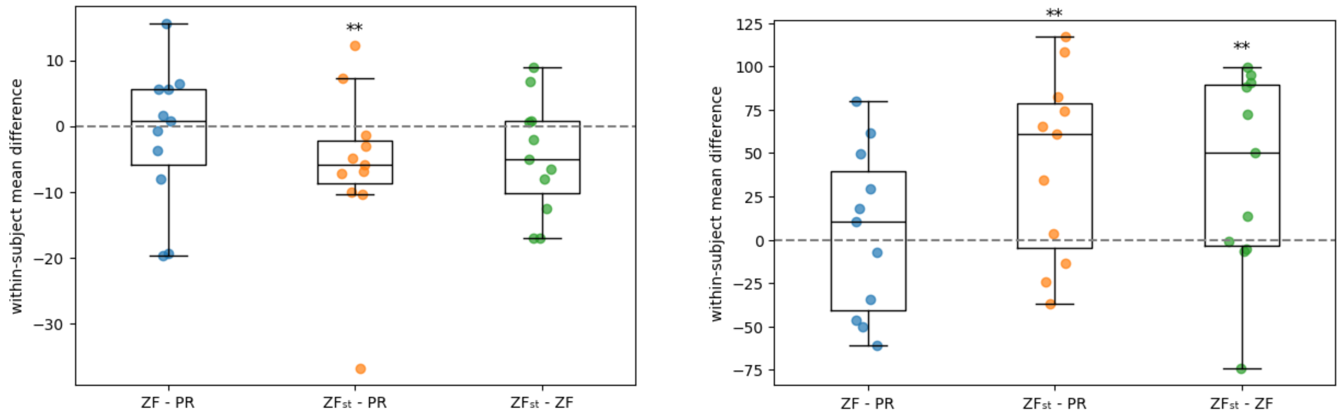
Figure 6: Within-subject mean differences for LSEQ subscales (ZF − PR, ZF_{ST} − PR, and ZF_{ST} − ZF) between different pairs of conditions. Each dot represents a participant; boxes show median and Inter Quarter Range. Positive values indicate better scores for the first condition. A dotted horizontal line at 0 marks the no-effect reference (no within-subject difference) between conditions. ** denotes a significant omnibus test ($p < .05$).

longer IBIs in both the ZenFlow conditions - the ZF_{ST} condition compared to PR ($\Delta = +41.23$ ms, $SE = 15.10$, 95% CI [5.45, 77.00], $p = .019$) and ZF ($\Delta = +40.11$ ms, $SE = 14.64$, 95% CI [5.42, 74.81], $p = .019$). Longer IBIs are generally associated with reduced physiological stress and greater autonomic recovery, suggesting that ZF_{ST} facilitated more restful early sleep compared to the other modalities. No significant difference was found between PR and ZF ($p = .997$). Together, these objective measures indicate that ZF_{ST} produced more favorable physiological outcomes in the pre-sleep period and during sleep onset.

Together, these objective measures indicate that ZF_{ST} produced more favorable physiological outcomes in the pre-sleep period and during sleep onset (Figure 7). This pattern aligns with the subjective findings, where ZF alone offered improvements relative to baseline in some subscales, but only with the addition of spatial transition (ZF_{ST}) did the intervention yield consistent and significant benefits across all subscales.

5.5.3 Qualitative Feedback. Through reflexive thematic analysis of twelve semi-structured interviews, we identified three themes that illustrate participants' experiences with traditional pranayama practices and technology-mediated interventions (ZF_{ST} and ZF). Together, these themes highlight both the potential of immersive systems to support stress management and sleep, as well as the challenges for long-term use.

Practical Barriers to Sustained Practice: Participants described challenges with both traditional pranayama and MR/AR when trying to integrate them into daily life. While pranayama was familiar, it was often limited by discomfort in posture and the effort needed to maintain focus. As one explained, (*"It was calming, but sitting upright and keeping posture was not very comfortable"* - P10). Another added, (*"You have to put your mind into it, which is harder when there is nothing else taking your attention"* - P11). These difficulties



(a) Pre-sleep stress scores. ZF_{ST} **showed lower stress than PR (**p = .032**).**

(b) Early sleep Inter-Beat Interval. ZF_{ST} **was higher than both PR and ZF (**p = .019**).**

Figure 7: Within-subject mean differences for objective measures from the Garmin Venu 3s smartwatch (ZF − PR, ZF_{ST} − PR, and ZF_{ST} − ZF) between different pairs of conditions. (a) Stress scores (lower the better) during the 30 minutes before sleep. (b) Mean IBI (higher the better) during the first 30 minutes of sleep. A dotted horizontal line at 0 marks the no-effect reference (no within-subject difference) between conditions. ** denotes a significant omnibus test (p < .05).

meant that benefits were often short-lived: (*"My heart rate slowed, I felt calmer, but it was hard to sustain"* - P05).

Similar concerns arose with ZF_{ST}/ZF. Some participants noted that headset use at bedtime required extra effort or was limited by device factors such as brightness and weight: (*"Even though ZF_{ST} was calming, putting on the headset before sleep feels like effort"* - P06). Yet, others emphasized that once engaged, ZF_{ST} felt easier to follow than pranayama because it offloaded the effort of self-regulation: (*"Even though the effort of taking the headset and putting it on is more, the activity itself feels easier"* - P11). These reflections highlight that both traditional and technological practices require attention to comfort and usability in order to be sustained.

Technology-Supported Relaxation: In contrast to the barriers of traditional practice, participants widely described ZF_{ST} and ZF as effective in helping them focus and relax. The structured guidance of visual and auditory cues was appreciated: (*"Visualizing what your breathing is supposed to feel like helps me focus more, because I tend to have trouble focusing"* - P11). The integration of music and visuals reduced effort, making sessions more engaging: (*"Listening to music, following the visuals, I didn't need to think of posture-it was more relaxing"* - P01).

ZF_{ST} in particular stood out for its immersive qualities. Several described the experience as (*"peaceful"* - P06), (*"dream-like"* - P02), or (*"helping me switch off"* - P07). One participant explained, (*"After ZF_{ST}, I was much more ready to sleep"* - P05). Others noted how the visuals created a sense of transport: (*"The night sky and stars were beautiful, it made me feel calm"* - P08). At the same time, not all experiences were uniformly positive. For example, one participant found the starry night imagery unsettling: (*"I personally don't associate night positively, sometimes it felt heavy"* - P10). Another reflected that although ZF_{ST} felt relaxing during use, old habits reduced the impact: (*"It does help when I am using it, very peaceful. But afterwards I continue my regular habit"* - P08). These varied accounts underline that while ZF_{ST} and ZF reliably produced in-the-moment calm, their longer-term effects depended on context and user habits.

Individual Preferences and Sustained Engagement: Most participants expressed a strong preference for ZF_{ST}, describing it as more immersive, calming, and effective in shifting attention away from stress. As one explained, (*"Definitely ZF_{ST}, I felt more relaxed and had better sleep"* - P02). Another described how (*"The visuals with stars and space background made me feel more relaxed"* - P07), while participant 11 highlighted how ZF_{ST} combined multiple cues seamlessly: (*"Seeing the stars grow brighter and dim with my breathing helped me focus, because I have trouble sticking to one thing"* - P11).

A few participants, however, favored other modalities. Some preferred ZF for its groundedness: (*"ZF felt bright, natural, and less intimidating, I felt safe"* - P09). Others appreciated pranayama for its simplicity and accessibility: (*"Pranayama is way easier because you don't need the stuff to do it"* - P11). These preferences underscore the role of personal choice in wellbeing practices.

Sustaining engagement was another shared concern. Several participants noted declining interest when exposed to the same content repeatedly: (*"First day very interesting, third day I already know the patterns. There are no surprises later"* - P08). To address this, participants recommended greater content variety, such as diverse natural environments and different music: (*"Adding more environments and different music would help sustain interest"* - P06). Taken together, these insights underline that ZF_{ST} was generally experienced as the most powerful and engaging modality, but personalization and variety remain key to ensuring long-term use.

5.5.4 Summary of Findings. Overall, our results strongly indicate that the ZF_{ST} condition, characterized by its careful design of virtual element together with seamless environmental transformation, was the most preferred and effective modality among the three tested. Both self-reported questionnaires and objective physiological data from wearable devices consistently demonstrated that ZF_{ST} experience led to significantly higher perceived relaxation, reduced stress levels, and improved sleep quality.

6 Discussion

Our findings reveal that the design of immersive relaxation experiences must consider not only the content but also the method of environmental transition. While both Zenflow variants—ZF (without spatial transition) and ZF_{ST} (with spatial transition)—provided improvements over traditional Pranayama, their effects varied across different sleep subscales. Specifically, ZF showed significant gains in some subscales (e.g., Quality of Sleep and Behavior Following Wakefulness), suggesting that the content alone can positively influence relaxation. However, only ZF_{ST} consistently outperformed Pranayama across all subscales, indicating that the addition of spatial transition enhances the overall efficacy of the intervention.

Interestingly, despite ZF_{ST}'s superior performance relative to baseline, statistical comparisons between ZF and ZF_{ST} did not yield significant differences. This suggests that spatial transition alone may not be a defining factor in relaxation outcomes. Instead, its value appears to emerge when paired with well-designed content—supporting a synergistic effect where perceptual continuity amplifies the impact of immersive cues.

These results underscore the importance of designing for both what users experience and how they enter that experience. Gradual transitions may scaffold attention and reduce cognitive load, enabling users to engage more deeply with the relaxation content. From a theoretical perspective, this aligns with embodied cognition and interoceptive awareness frameworks, which emphasize the role of environmental coherence in emotional regulation.

Overall, our results indicate that the ZF_{ST} condition—combining carefully designed virtual elements with seamless environmental transitions—was the most effective and preferred modality among the three tested. Both subjective self-reports and objective physiological measures consistently showed that ZF_{ST} produced greater perceived relaxation, lower stress levels, and enhanced early sleep quality compared to the other conditions.

These insights have broader implications for the design of XR-based therapeutic systems. Rather than relying solely on immersive content or abrupt transitions, future systems should consider how spatial transformation can be integrated meaningfully to support relaxation. Personalization, content variety, and adaptive feedback mechanisms may further enhance long-term engagement and efficacy.

Personalization and Sustained Engagement: Despite ZF_{ST}'s effectiveness, participants highlighted the need for personalization and content variety. Repeated exposure to the same environment led to habituation, reducing engagement over time. This points to future opportunities for adaptive MR systems that dynamically adjust visuals, soundscapes, and pacing based on user preferences or biofeedback. Designing for sustained engagement may require rotating environments, integrating seasonal themes, or allowing users to curate their own relaxation spaces.

Implications for XR Therapeutics: Zenflow contributes to the growing field of digital therapeutics by demonstrating that environmental transition design can significantly impact relaxation efficacy. The use of consumer-grade wearables for physiological validation also highlights a scalable pathway for real-world deployment. Importantly, the co-design process with clinical experts ensured that the system avoided common pitfalls—such as triggering imagery or overly directive agents—and remained aligned with therapeutic principles. This underscores the value of participatory design in developing safe and effective XR interventions.

Limitations and Future Work: Our study involved a small sample (N=12) and a short intervention period (three weeks), limiting generalizability. Future research should involve larger, more diverse populations and explore long-term adherence. While Garmin wearables provided useful data, clinical-grade sensors could offer more precise insights into autonomic regulation. Furthermore, the study's duration of three weeks, with 8-minute daily sessions, provides a snapshot of short-term effects; longitudinal studies are needed to assess the sustained impact of these relaxation modalities.

We also plan to investigate multimodal transitions—incorporating haptic and auditory cues—and explore personalized MR environments that respond to real-time biofeedback. Clinical trials with populations experiencing chronic stress or sleep disorders could further validate Zenflow 's therapeutic potential.

7 Conclusion

This work presents Zenflow, a Mixed Reality system designed to support relaxation and sleep through gradual environmental transformation. Developed through co-design with clinical experts, Zenflow explores how perceptual continuity and spatial transitions can enhance the efficacy of immersive interventions. Our three-week within-subjects study compared traditional Pranayama with two Zenflow variants—one without spatial transition (ZF) and one with spatial transition (ZF_{ST}). Results from both subjective questionnaires and physiological data reveal that while ZF offered modest improvements over baseline, ZF_{ST} consistently outperformed both ZF and Pranayama across sleep quality and stress reduction metrics.

These findings underscore the importance of transition design in MR experiences. Rather than abrupt immersion, gradual spatial transformation—paired with subtle breathing cues—can scaffold users into deeper states of relaxation. In sum, MR experience and spatial transitions by themselves offered limited gains, but when combined experience with spatial transformation ZF_{ST}, the intervention delivered clear and significant improvements over baseline.

Looking forward, Zenflow opens new directions for adaptive MR therapeutics that respond to user physiology and preferences. As immersive technologies become more accessible, designing for perceptual continuity and therapeutic appropriateness will be key to supporting mental well-being in everyday environments.

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